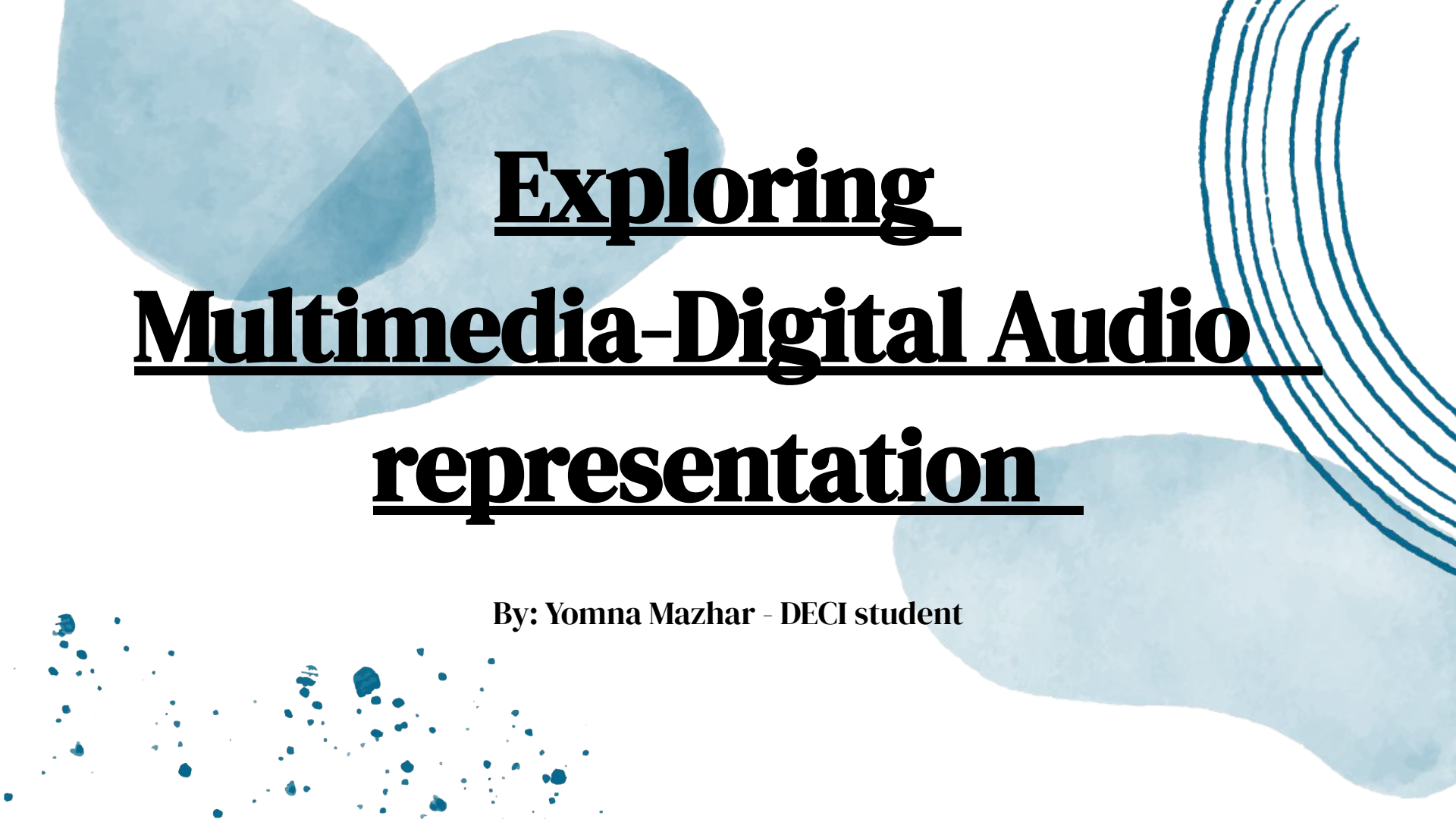
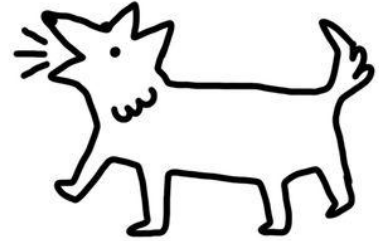




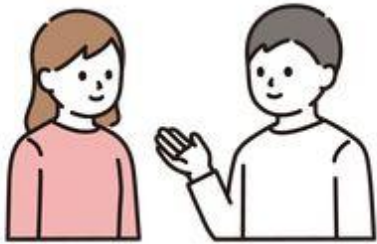
# **Exploring** **Multimedia-Digital Audio** **representation**

By: Yomna Mazhar - DECI student

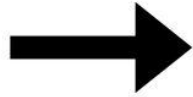
**All that we hear like animals sounds , our taking  
or music are invisible sound waves in the air.**



**These sound waves become visible when we represent them as a files digitally on our computers and become digital audio.**



**Frist: recording or capture  
Sound waves.**



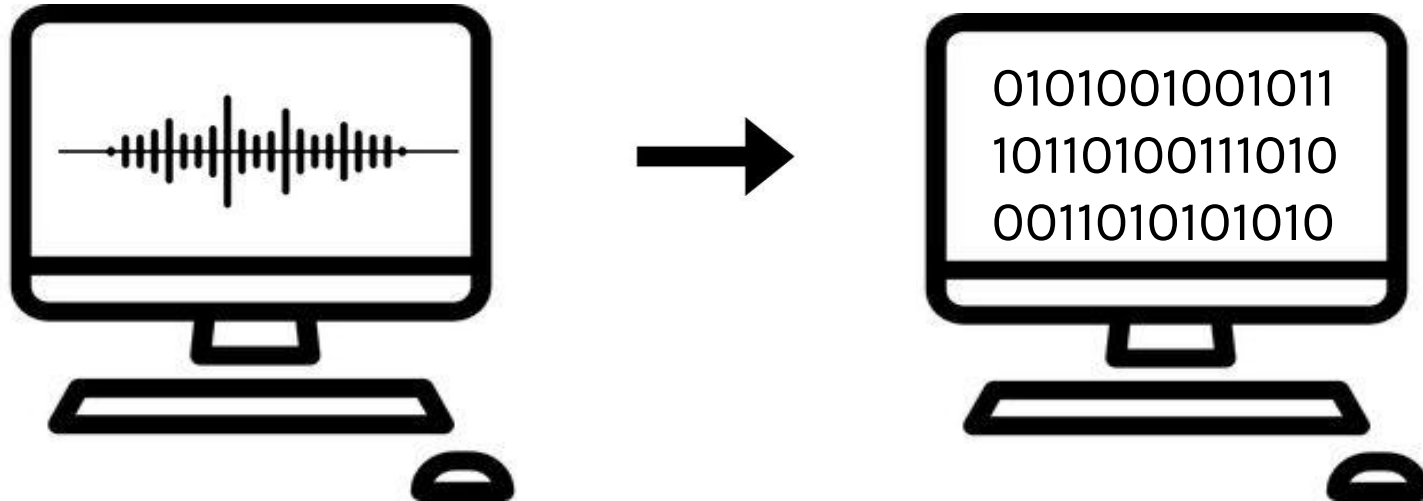
**Then save these waves  
on the computer as a  
file.**



**Sound waves changes into visible  
waves by representing digitally 'the  
digital audios'.**

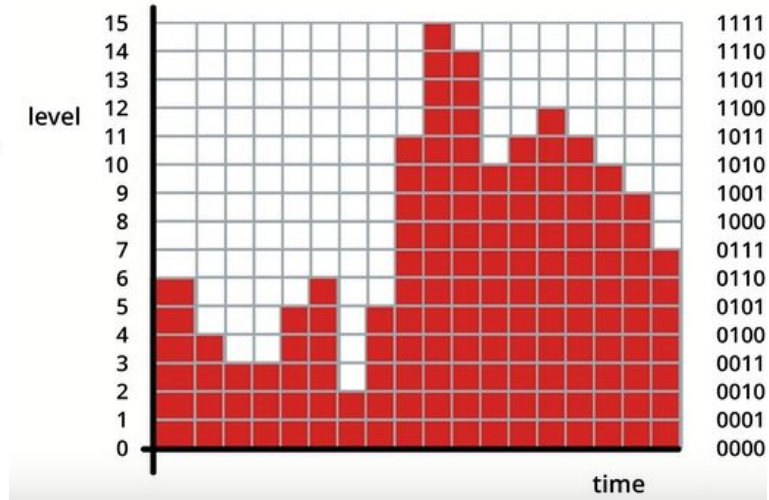
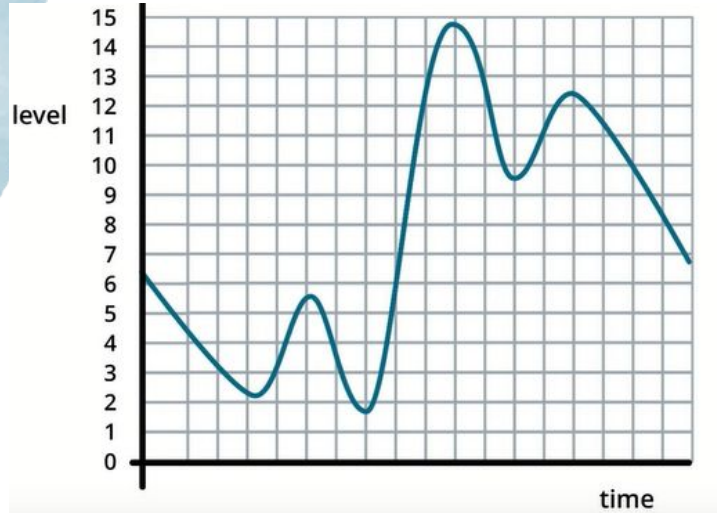
**The representation of sound waves on computer are analog signals, which means they are continuous and vary in amplitude and frequency.**

**The ability of converting the physical sound waves into zeros and ones “binary ” is called analog to digital conversion ‘ADC’.**

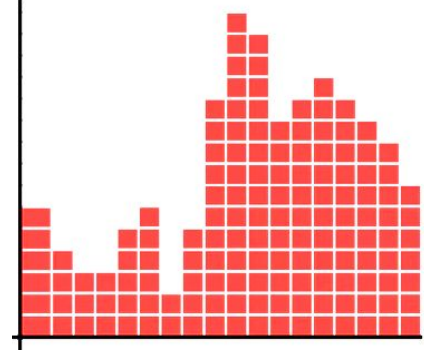


# how does the computer microphones convert sound into binary data?

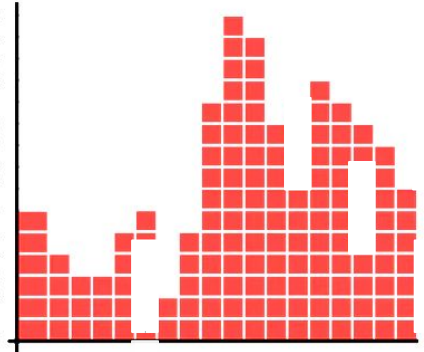
The sound has up and downs which splits into a graph which converted into samples each sample has a decimal number which converted to binary by digital converter ... these ups and downs called “digital waves”



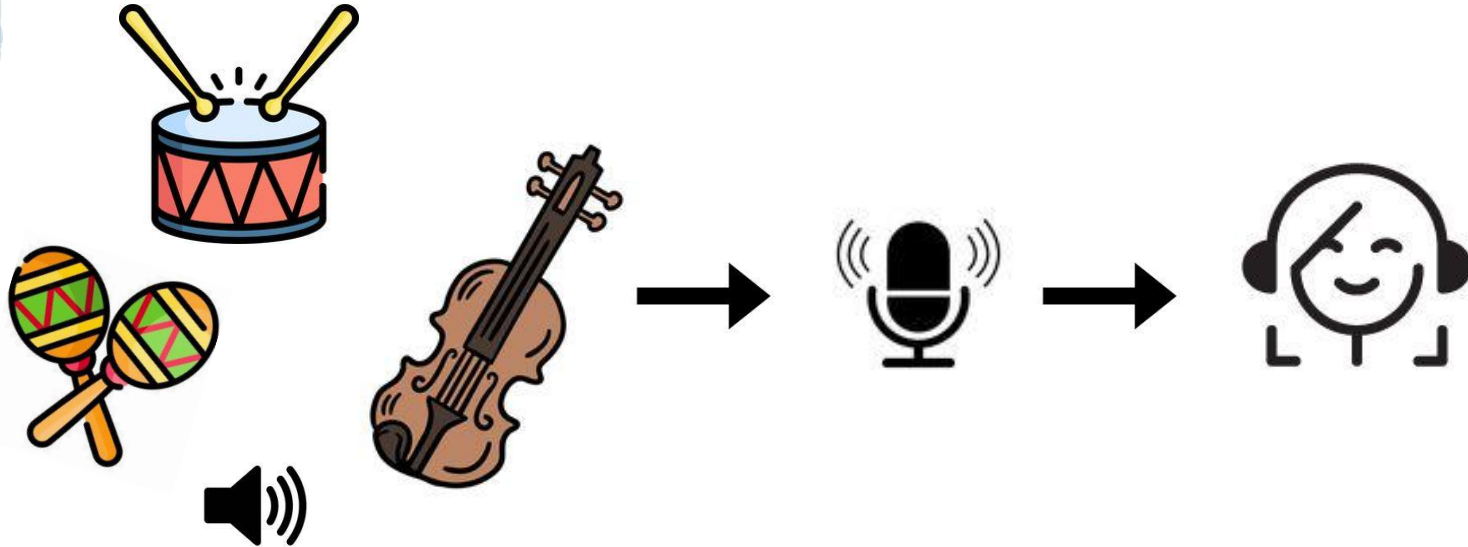
**Digital waves can be represented in digital data very accurately if a digital converter takes millions of samples and the sound will be high quality.**



**If the digital converter takes a few samples the audio will sound bad and poorly because of losing some data of the physical sound waves.**



**When information from the world is captured and made digital, like sound waves, capturing as much information as possible results in the best quality information and the best sound.**

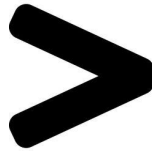




**Very high-quality digital audio and video data is  
has too much of samples and simply too much  
information to store or stream and vise versa.**



**10 MB  
more quality  
More samples  
More space**



**5 MB  
Less quality  
Less samples  
Less space**



# So we have to compress data according to the purpose or what we will use for and how it is into file formats

## Compression classified into :

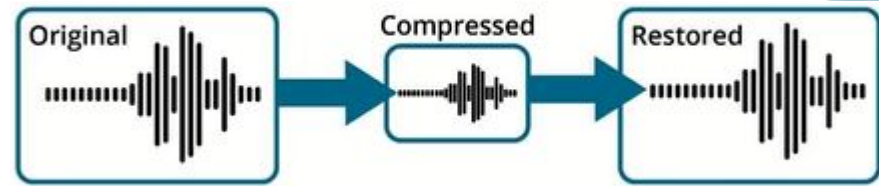
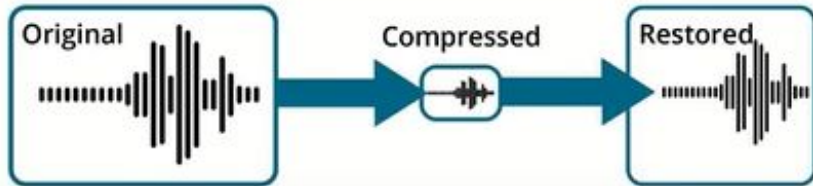
### **Lossy compression:**

It is compression data by losing some information that cannot be restored “ reduce space ”



### **Lossless compression:**

It is compression the data without losing information “ do not reduce space ”



# Digital Audio Formats

## Lossless :

- **FLAC (.flac)**
- **WAV (.wav)**
- **AIFF (.aiff)**
- **DSF (.dsf)**

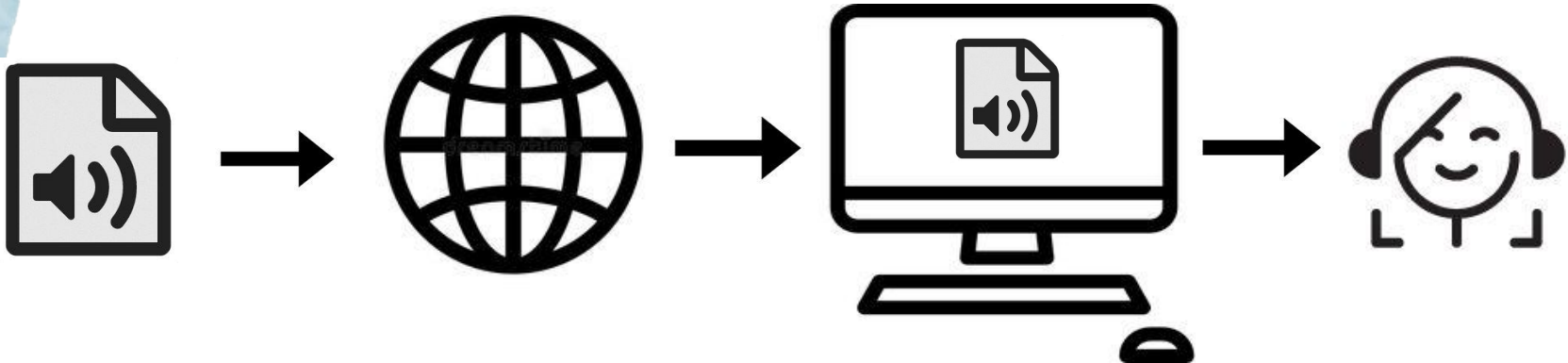


## Lossy :

- **MP3 (.mp3)**
- **AAC (.aac)**
- **OGG (.ogg)**
- **WMA (.wma)**



**"streaming" the audio data means that The audio and video files are broken up and sent over the internet in small chunks, and arrive at your computer or device as you are listening**



Code on servers and the web browser, and in mobile platforms sends or receives larger audio and video files in pieces over the internet.

**bit rate** : the of of information being sent over the internet how many bits per second are being sent from a server computer to your computer (bit per sec).by increasing the speed of internet the bit rate increases

**bit depth** :The amount of information in one of these streams .



The background features several abstract blue watercolor shapes. In the top left, there are two overlapping, soft-edged blue blobs. In the bottom right, there is a larger, lighter blue shape. A series of thin, dark blue curved lines sweep across the right side of the image. In the bottom left corner, there is a cluster of small, dark blue dots and speckles.

**Thanks a lot !**

**By: yomna mazhar**